Global Health Economy Governance Dynamics (1960-2024): A Visual Exploration of Long-Term Development Patterns

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ABSTRACT

In 1960, a child born in Oman could expect to live just 35 years; by 2024, that number had risen to 80. This project used interactive visualizations to explore the relationship between economic growth, democratic governance, and health outcomes across 236 countries over six decades. Integrating Gapminder/World Bank health-economic data with V-Dem governance indices, we found a logarithmic relationship between GDP and life expectancy with diminishing returns, a positive but non-deterministic association between democracy and health, and evidence that the largest gains occurred in nations that started poorest, regardless of regime type.

1. Introduction and Background

In 2006, Hans Rosling stood on a TED stage and shattered a room full of assumptions. Using animated bubble charts, he showed that the world was not divided into a "Developed" West and an "Undeveloped" rest; that countries like Bangladesh and Vietnam had achieved health outcomes rivaling wealthy nations at a fraction of the income (Rosling et al., 2018). His visualization did what tables of numbers could not: it made six decades of global progress visible, Intuitive, and undeniable. Nearly two decades later, the questions Rosling raised remain urgent. Has economic growth consistently delivered longer lives, or have diminishing returns set in? Do democratic institutions independently improve health, or is wealth the only thing that matters? And when crises like HIV/AIDS and COVID-19 strike, which nations prove resilient and which collapse?

This project tackled these questions using the lens of data visualization. Following Munzner's (2014) nested model for visualization design, we begin at the **domain level,** the real-world context of global development policy, where decisions about resource allocation, governance reform, and health investment affect billions of lives. At the **data abstraction level**, we integrate two datasets: a health-economic dataset combining Gapminder and World Bank indicators (Gapminder Foundation, 2024; World Bank, 2024) with a governance dataset from the Varieties of Democracy project (Coppedge et al., 2025). At the **idiom level**, we design six visualizations: maps, bar charts, line charts, scatter plots, tables, and heat maps, each chosen to match the perceptual demands of the task. A choropleth map leverages spatial position, the strongest visual channel, to provide geographic overview (Munzner, 2014, Ch. 5). Grouped bar charts use aligned length on a common baseline, avoiding the well-documented weakness of stacked bars where middle segments lack a shared comparison point (Munzner, 2014, Ch. 7). A Gapminder-inspired bubble chart uses a logarithmic scale to spread low-income nations that would otherwise be compressed at the origin, while encoding population as bubble size and region as color hue, a categorical channel well-suited to distinguishing unordered groups (Munzner, 2014, Ch. 5).

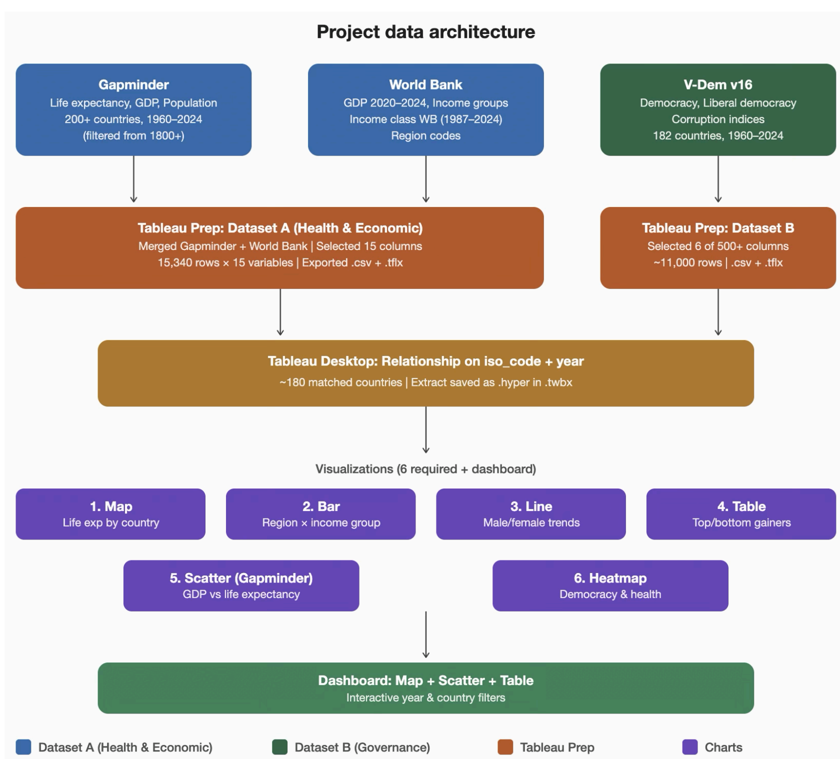
1.2 Research Questions

1. Has economic growth consistently translated to longer lives?
2. Do democratic institutions improve health outcomes at similar income levels?
3. Are world regions converging or diverging in development?
4. Which countries achieved the greatest life expectancy gains, and what do they share?
5. How did major crises (COVID-19, HIV/AIDS) affect different country groups?

The project followed Shneiderman's (1996) information-seeking mantra: "overview first, zoom and filter, then details on demand", which Munzner endorses as a foundational design principle (Munzner, 2014, Ch. 6). Our interactive dashboard provides the overview; year sliders and country filters enable zooming; and tooltips deliver details on demand. Understanding which factors, wealth, governance, corruption, and policy, drive health improvements is fundamental to development economics and public health strategy. If democratic institutions independently improve health outcomes even after controlling for income, then governance reform deserves as much attention as economic growth in development agendas. If the largest gains come from escaping extreme poverty, then resources should be targeted at the poorest nations.

2. Datasets

This project integrated two datasets prepared in Tableau Prep and connected through a relationship in Tableau Desktop.



2.1 Dataset A: Health and Economic Indicators

Dataset A was constructed in Tableau Prep by merging data from two sources. From the Gapminder Foundation: a Swedish non-profit that compiles global development statistics (Gapminder Foundation, 2024), we selected life expectancy for male and female populations, GDP per capita, and population, filtering the data to the period 1960–2024. An overall life expectancy variable was derived by averaging the male and female values, a computed attribute. From the World Bank World Development Indicators (World Bank, 2024), we added GDP data for 2020–2024 to compensate for the gap where Gapminder coverage ended around 2019, along with income group classifications, the historical Income Class WB variable (available from 1987 to 2024), and regional codes. In the Tableau Prep flow, we selected 15 relevant columns, merged the two sources on country and year, and exported both the cleaned CSV and the Prep flow file (.tflx).

We classified this as a tabular dataset where each row (item) represented a country-year observation. Key attributes included: life expectancy, calculated field overall, male, and female (quantitative, sequential); GDP per capita in PPP dollars (quantitative, sequential); population (quantitative, sequential); region at two granularities; 4-region and 6-region (categorical, nominal); income group, the current World Bank classification (ordinal); and income class WB; a historical classification tracking how countries moved between income levels from 1987 to 2024 (ordinal, temporal). Geographic coordinates enabled spatial encoding on maps.

Limitations included: GDP per capita combined two sources with possible minor methodological differences at the 2019–2020 boundary; the historical income classification began only in 1987, producing null values for earlier years; and some small territories lacked complete data.

2.2 Dataset B: Governance Indicators

Dataset B drew from the Varieties of Democracy (V-Dem) project version 16, hosted by the University of Gothenburg, which employs over 3,700 country experts to code democratic indicators (Coppedge et al., 2025). In Tableau Prep, we selected 6 columns from V-Dem's 2000 indicators, focusing on the three most relevant to our research questions. The cleaned dataset contained approximately 11,000 rows covering 182 countries. Three governance indicators were retained, all quantitative and scaled 0 to 1: the Electoral Democracy Index (V2X Polyarchy), measuring free elections and civil liberties; the Liberal Democracy Index (V2X Libdem), extending to constitutional protections and judicial independence; and the Political Corruption Index (V2X Corr), measuring corruption across government branches, where higher values indicated more corruption.

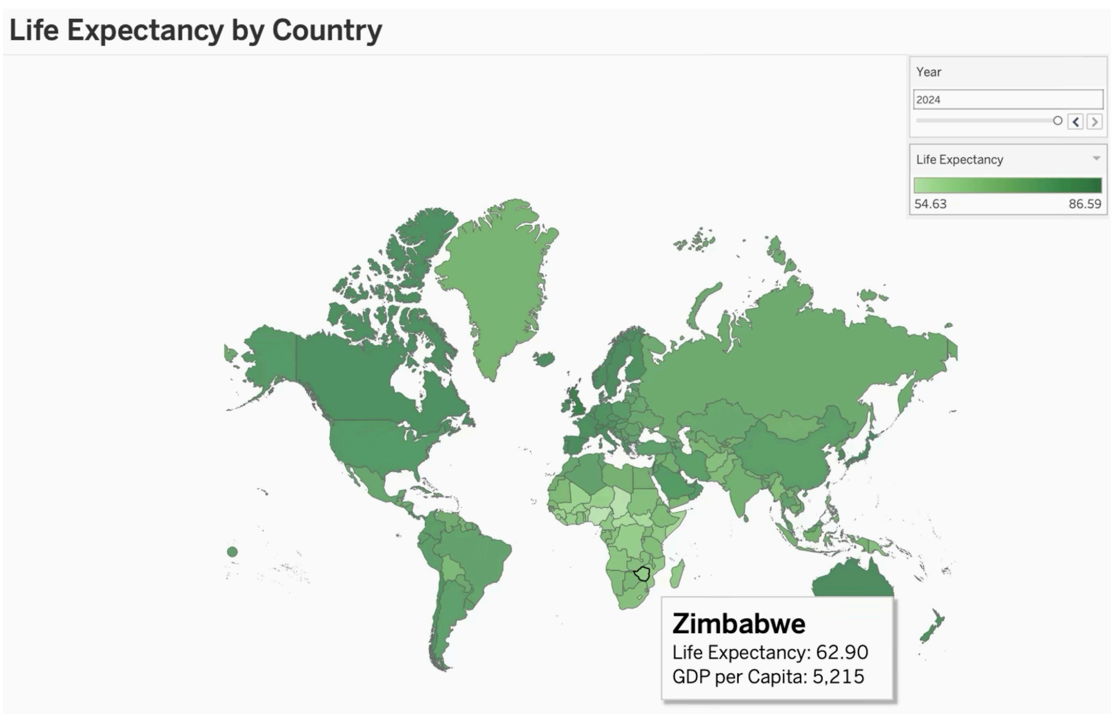
Limitations included: V-Dem covered 182 countries versus 236 in Dataset A, leaving approximately 54 smaller territories without governance data. Expert-coded indices involved subjective judgment, though V-Dem mitigated this through multiple coders per country.

Integration in Tableau Desktop
The two datasets were connected in Tableau Desktop using a relationship on ISO country code and year, not a physical join. Following Tableau's recommended approach, relationships preserved each table's level of detail and allowed Tableau to determine the appropriate join at query time, avoiding measure duplication that can occur with joins in one-to-many scenarios. The matched dataset covered approximately 180 countries where both health-economic and governance data were available. The connection was saved as a packaged extract (.hyper) within the .twbx workbook file.

3. Visual Analysis and Findings
Our visual analysis followed Shneiderman's (1996) mantra: overview first, then zoom and filter, then details on demand. We began with a geographic overview, moved to regional and temporal comparisons, and then drilled into country-level patterns.

3.1 Geographic Overview: Life Expectancy by Country

Where Does the World Stand? (Q3)

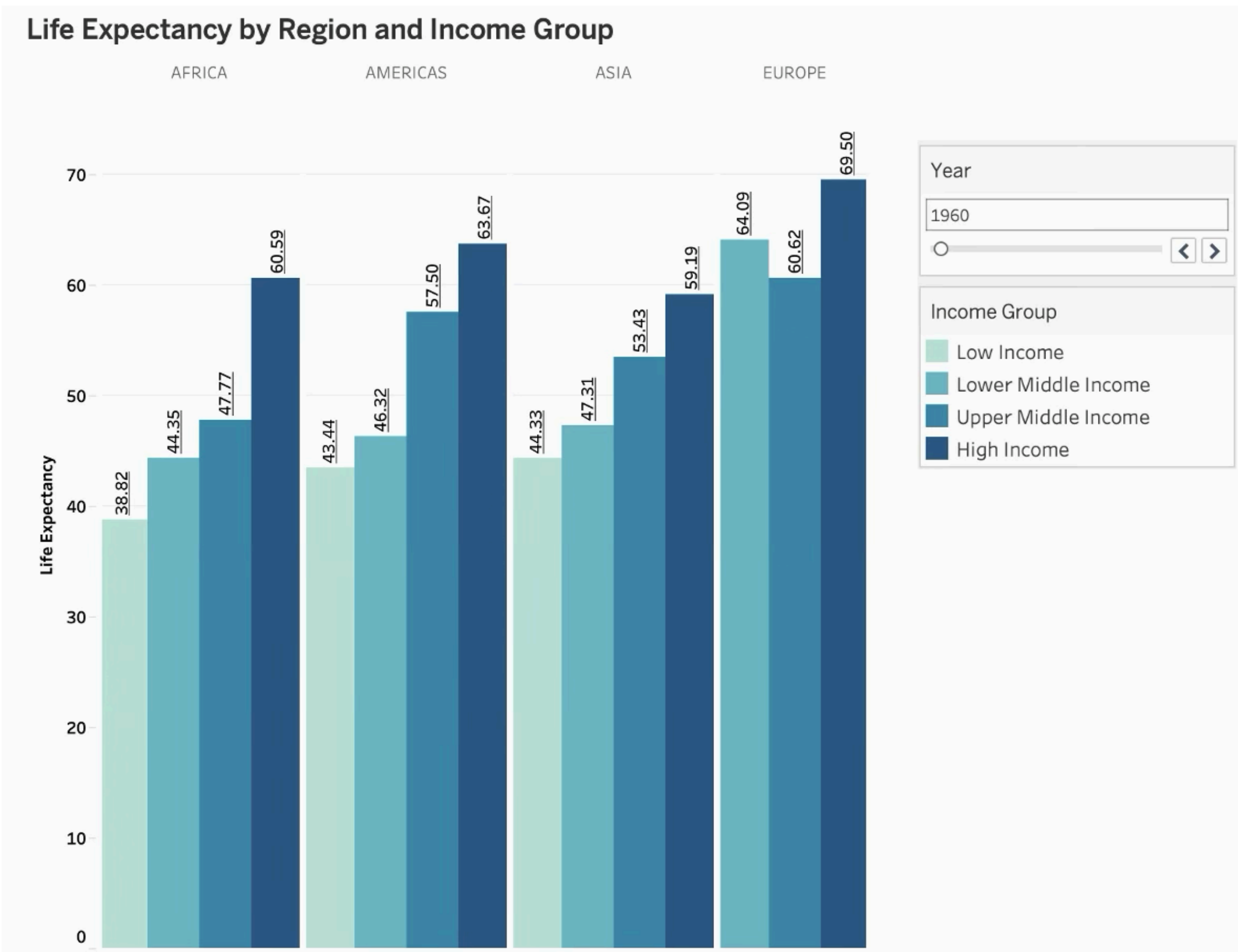


Data source: Dataset A. Variables: life_expectancy (quantitative, color), country_name (categorical, geographic mark), gdp_per_capita (quantitative, tooltip), year (ordinal, filter).

This choropleth map provided a geographic overview of life expectancy across the globe for a selected year. Countries were shaded using a sequential green palette from light (approximately 54 years) to dark (approximately 87 years), with color luminance encoding quantitative magnitude, a channel ranking supported by Cleveland & McGill's (1984) seminal graphical perception experiments and later formalized in automated visualization systems (Munzner, 2014). Each country mark encodes life expectancy by color and GDP per capita in the tooltip, enabling details on demand. An interactive year slider allowed readers to observe changes from 1960 to 2024. The map immediately revealed the global health divide: Sub-Saharan Africa appeared distinctly lighter than Europe, East Asia, and the Americas. Sliding from 1960 to 2024 showed progressive darkening across most regions, illustrating remarkable global progress while highlighting regions that lagged.

3.2 Regional Comparison: Life Expectancy by Region and Income Group

Are Regions Converging or Diverging? (Q3)



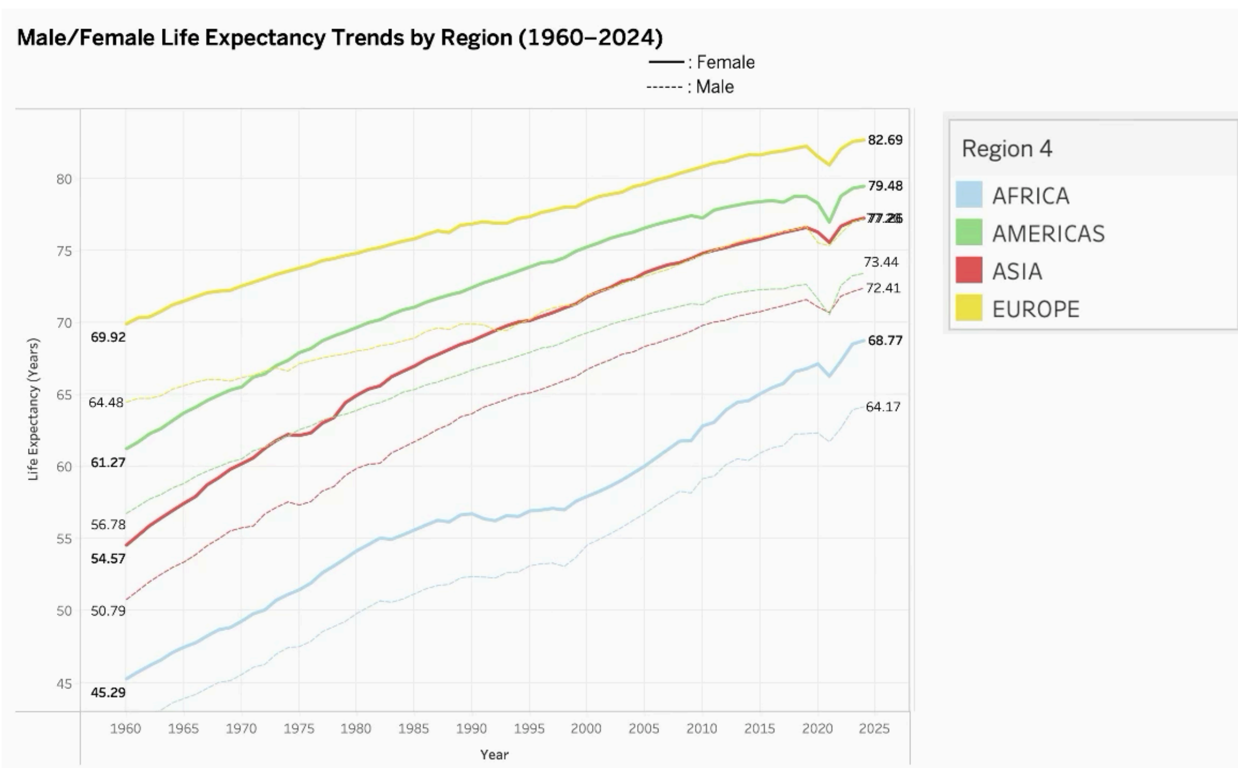
Data source: Dataset A. Variables: region_4 (categorical, position), income_group (ordinal, color + position), life_expectancy (quantitative, length), year (ordinal, filter).

This grouped bar chart compares average life expectancy across four world regions (Africa, the Americas, Asia, and Europe), with bars further broken down by World Bank income classification. We used grouped bars rather than stacked bars because only the bottom segment of a stacked bar shares a common baseline, making middle and top segments difficult to compare across categories (Franconeri et al., 2021). With grouped bars, every bar originated from zero, enabling accurate length-based comparison across all income groups. The chart used a sequential blue-teal palette where lighter shades represented lower income and darker shades represented higher income groups, following luminance-based encoding guidelines for ordered data. Value labels on each bar provided precise readings without requiring the reader to estimate from the axis.

The chart addressed Q3 (convergence/divergence) by revealing that within every region, higher income groups achieved higher life expectancy, but the magnitude of the gap varied. When the reader slid from 1960 to 2024, the income-based gap narrowed within each region, evidence of convergence consistent with the WHO's finding that global life expectancy increased by over 6 years between 2000 and 2019 before COVID-19 disrupted progress (World Health Organization, 2022).

3.3 Temporal Trends: Male/Female Life Expectancy by Region (1960–2024)

How Did Crises and Gender Shape Progress? (Q5)



Data source: Dataset A. Variables: year (quantitative, position-x), life_expectancy_female and life_expectancy_male (quantitative, position-y), region_4 (categorical, color hue).

This dual-axis line chart tracked average life expectancy over six decades for four world regions, with separate lines for female (solid) and male (dashed) populations. Color hue distinguished regions, an appropriate channel for unordered categorical data (Cleveland & McGill, 1984), while line style (solid vs. dashed) distinguished gender, using a shape channel that remained discernible even without color. End-of-line labels displayed the 2024 values, reducing the need for back-and-forth legend lookup, an application of the "eyes beat memory" principle where information needed for comparison should be visible simultaneously rather than stored in working memory.

This chart addressed Q5 (crisis impacts) and the gender gap. The upward trajectory across all regions confirmed remarkable global progress. However, two crisis events were clearly visible: Africa's lines flattened and dipped during the 1990s–2000s, reflecting the HIV/AIDS epidemic that reversed decades of health gains, a pattern well-documented by UNAIDS, which estimated that life expectancy in the hardest-hit Southern African countries fell by over 10 years between 1990 and 2005 (UNAIDS, 2024). Around 2020, all regional lines showed noticeable dips corresponding to the COVID-19 pandemic, which the WHO estimated erased nearly a decade of progress in healthy longevity globally (World Health Organization, 2022). The persistent gap between solid and dashed lines, typically 3–5 years, with women outliving men across all regions, revealed a universal gender disparity consistent with global demographic research attributing the gap to biological, behavioral, and occupational factors (GBD 2021 Causes of Death Collaborators, 2024).

3.4 Country Gains: Top/Bottom Life Expectancy Gains (1960–2024)

Who Improved Most and What Do They Share? (Q4)

Top/Bottom Countries: Life Expectancy Gains (1960–2024)

In / Out of Top 10 Gainers	Country Name	Life Exp 1960	Life Exp 2024	Life Exp % Gain	GDP Per Capita 1960	GDP Per Capita 2024	GDP Per Capita % Gain	Democracy %	Corr % change
Top 10 Gainers	China	33.51	78.11	133.07%	$1,155.00	$23,845.62	1964.56%	-22.34%	0.18%
	Oman	35.54	80.38	126.20%	$4,228.00	$36,721.08	768.52%	4.79%	-0.78%
	Maldives	38.26	81.47	112.93%	$1,062.00	$23,034.41	2068.97%	-3.83%	-32.29%
	Timor-Leste	32.45	67.97	109.45%	$734.50	$3,890.96	429.74%	16.14%	-19.05%
	Algeria	41.04	76.50	86.41%	$6,759.00	$15,501.92	129.35%	-19.25%	-1.19%
	Nepal	38.61	70.61	82.87%	$1,066.00	$5,046.82	373.43%	20.32%	-13.13%
	Morocco	41.73	75.56	81.09%	$1,997.00	$9,162.75	358.83%	-0.38%	-4.05%
	Iran	43.28	77.91	80.02%	$4,907.00	$17,484.13	256.31%	14.48%	6.39%
	Saudi Arabia	45.45	79.33	74.56%	$9,268.00	$62,792.77	577.52%	-25.00%	-30.49%
	Tunisia	44.06	76.73	74.15%	$2,214.00	$12,774.92	477.01%	28.66%	22.39%
Bottom 10 Gainers	Lesotho	49.73	57.74	16.11%	$567.40	$2,640.29	365.33%	13.59%	-2.81%
	Jamaica	63.74	71.61	12.35%	$6,226.00	$11,340.26	82.14%	4.04%	12.43%
	Slovak Rep..	70.32	78.46	11.57%	$9,036.00	$40,319.05	346.20%	-12.34%	-5.76%
	Bulgaria	69.19	75.85	9.62%	$5,010.00	$34,221.53	583.06%	-4.02%	13.31%
	Latvia	69.90	76.18	8.99%	$5,597.00	$37,611.67	572.00%	2.21%	-39.83%
	Lithuania	70.01	76.11	8.71%	$9,466.00	$47,165.37	398.26%	-5.79%	17.36%
	Russia	67.63	73.33	8.44%	$10,570.00	$41,704.71	294.56%	-42.81%	8.02%
	Ukraine	68.95	74.59	8.18%	$5,637.00	$16,319.84	189.51%	-19.88%	-29.04%
	Belarus	68.91	74.48	8.08%	$4,046.00	$29,040.97	617.77%	-25.93%	16.71%
	Nauru	58.42	62.30	6.64%	$16,090.00	$12,468.76	-22.51%		

Data sources: Dataset A (life_expectancy, gdp_per_capita) + Dataset B (v2x_polyarchy, v2x_corr). Derived attributes: Life Exp % Gain, GDP % Gain, Democracy % Change, Corruption % Change (all FIXED LOD calculations).

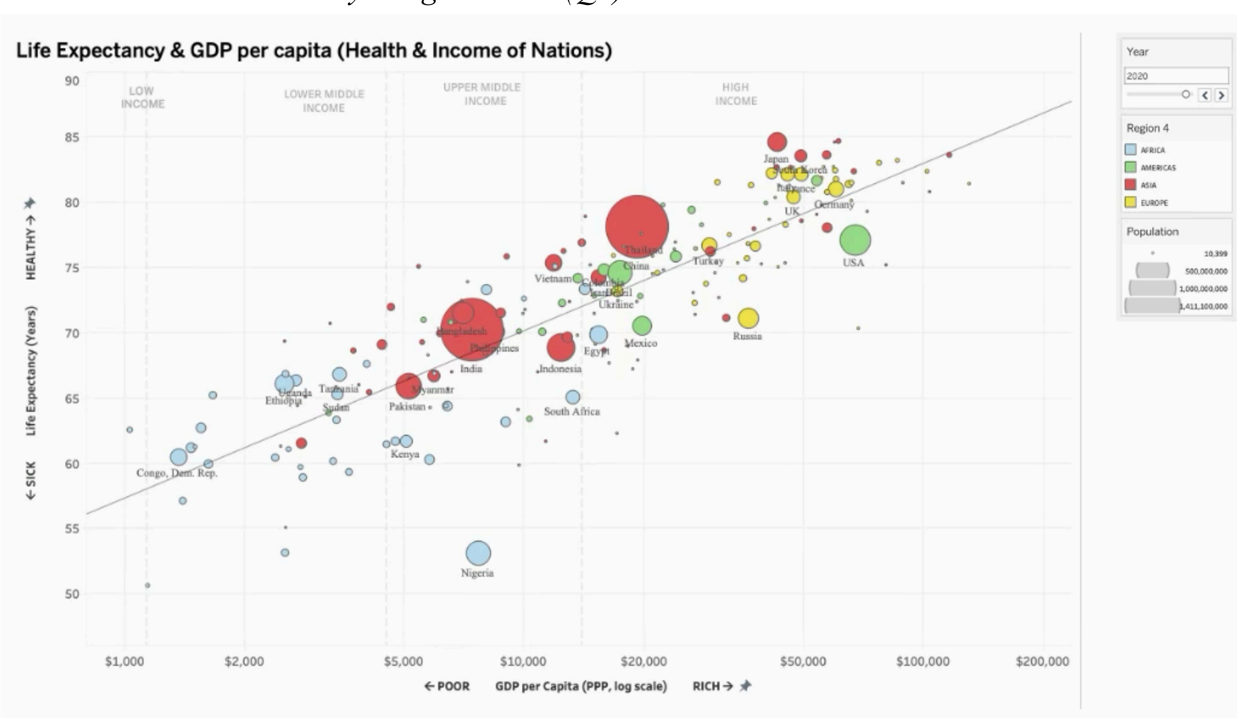
This text table compared the 20 countries with the highest and lowest life expectancy percentage gains from 1960 to 2024. Columns included starting life expectancy (1960), ending life expectancy (2024), percentage gain, GDP per capita in 1960 and 2024 with percentage gain, democracy percentage change, and corruption percentage change. We used FIXED Level of Detail calculations in Tableau to ensure the gain values remained constant regardless of the year filter, a technique that computed values across the full time range while allowing other columns to respond dynamically to the interactive year slider.

The top gainers were striking: China achieved a 133% life expectancy gain (from 33.5 to 78.1 years), Oman 126% (from 35.5 to 80.4), and Maldives 113%. These countries shared extremely low starting points in 1960 but varied widely in governance. China's democracy score changed by −22% while Nepal's increased by 20%. Their GDP gains were equally dramatic, with Oman's GDP increasing by 769% and Maldives by 2,069%. This pattern aligned with recent convergence research showing that poorer countries with basic institutional capacity tend to close health gaps faster than wealthier nations (Mutlu et al., 2024; (Ndzignat Mouteyica & Ngepah, 2024).

The bottom 10 gainers told an equally important story: Belarus (8.1%), Ukraine (8.2%), and Russia (8.4%) all started with relatively high life expectancy in 1960 (67–70 years) and barely improved. These were former Soviet states where political transition, economic disruption, and public health crises stalled progress, a stagnation pattern that persists in post-Soviet health systems to this day (OECD, 2023). Several also showed negative democracy percentage changes, suggesting that institutional decline accompanied health stagnation. The corruption data added further nuance: Latvia's corruption decreased by 39.8% alongside modest health gains, while Russia's corruption increased by 8%, accompanying near-zero health improvement.

3.5 Wealth and Health: GDP per Capita vs Life Expectancy

Does Economic Growth Buy Longer Lives? (Q1)



Data source: Dataset A. Variables: gdp_per_capita (quantitative, position-x, log scale), life_expectancy (quantitative, position-y), population (quantitative, area), region_4 (categorical, color hue), country_name (categorical, label/detail).

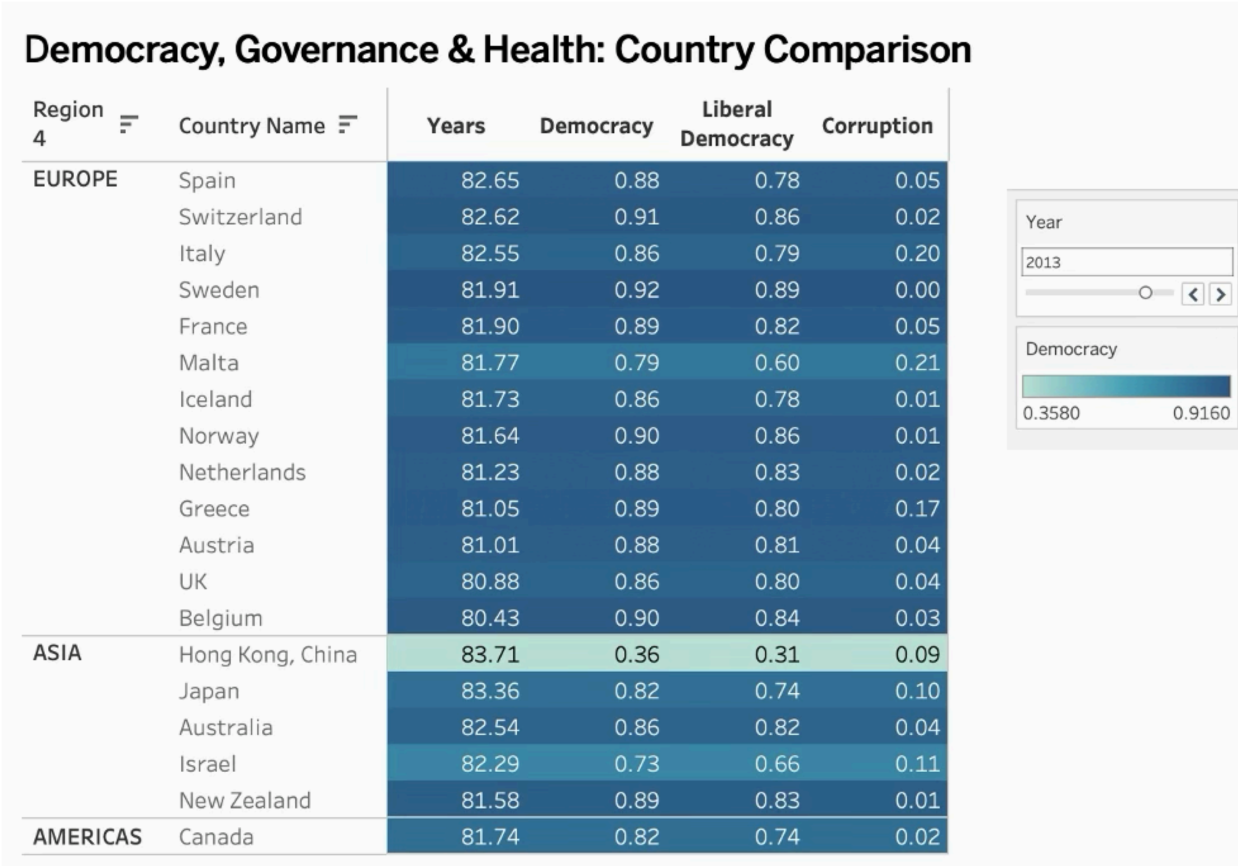
Inspired by Hans Rosling's iconic Gapminder visualization (Rosling et al., 2018), this bubble chart plotted GDP per capita against life expectancy for all countries. Following the channel effectiveness ranking where position is the strongest quantitative channel (Franconeri et al., 2021), we mapped GDP and life expectancy to horizontal and vertical position respectively. Population was encoded as bubble area, and region as color hue. The x-axis used a logarithmic scale, which spread out low-income countries that would otherwise have been compressed at the origin, a critical design choice given that most of the world's population lived below $10,000 GDP per capita. Income level annotations at the top of the chart provided contextual thresholds following World Bank classifications, with vertical dashed reference lines marking the boundaries. Axis labels included directional cues (← POOR / RICH → and ← SICK / HEALTHY →) inspired by the original Gapminder design. Only countries with populations exceeding 100 million were labeled directly to reduce clutter while highlighting the most significant nations.

This chart addressed Q1 (has economic growth translated to longer lives?). The logarithmic trend line confirmed a strong positive relationship consistent with the Preston Curve; the well-established finding that income gains yield large health improvements at low income levels but diminishing returns at higher levels (de la Escosura, 2023). Moving from $1,000 to $5,000 GDP per capita was associated with roughly 15 additional years of life expectancy, while moving from $50,000 to $100,000 yielded almost no additional gain. Notable outliers included the USA, which despite having among the highest GDP, achieved lower life expectancy than many European and Asian countries with half its income, a pattern that recent research attributes to inequality, healthcare access barriers, and lifestyle factors (GBD 2021 Collaborators, 2024). Conversely, Vietnam and Bangladesh achieved life expectancy above 70 years despite GDP per capita below $5,000, suggesting that effective public health policy can compensate for limited wealth.

Governance and Health: Democracy, Liberal Democracy, and Corruption *Do Democratic Institutions Matter? (Q2)*

3.6 Governance and Health: Democracy, Liberal Democracy, and Corruption

Do Democratic Institutions Matter? (Q2)



Region 4	Country Name	Years	Democracy	Liberal Democracy	Corruption
EUROPE	Spain	82.65	0.88	0.78	0.05
	Switzerland	82.62	0.91	0.86	0.02
	Italy	82.55	0.86	0.79	0.20
	Sweden	81.91	0.92	0.89	0.00
	France	81.90	0.89	0.82	0.05
	Malta	81.77	0.79	0.60	0.21
	Iceland	81.73	0.86	0.78	0.01
	Norway	81.64	0.90	0.86	0.01
	Netherlands	81.23	0.88	0.83	0.02
	Greece	81.05	0.89	0.80	0.17
	Austria	81.01	0.88	0.81	0.04
	UK	80.88	0.86	0.80	0.04
	Belgium	80.43	0.90	0.84	0.03
ASIA	Hong Kong, China	83.71	0.36	0.31	0.09
	Japan	83.36	0.82	0.74	0.10
	Australia	82.54	0.86	0.82	0.04
	Israel	82.29	0.73	0.66	0.11
	New Zealand	81.58	0.89	0.83	0.01
AMERICAS	Canada	81.74	0.82	0.74	0.02

Data sources: Dataset A (life_expectancy, region_4) + Dataset B (v2x_polyarchy, v2x_libdem, v2x_corr). Variables: country_name (categorical, rows), governance indices (quantitative, text + color), year (ordinal, filter).

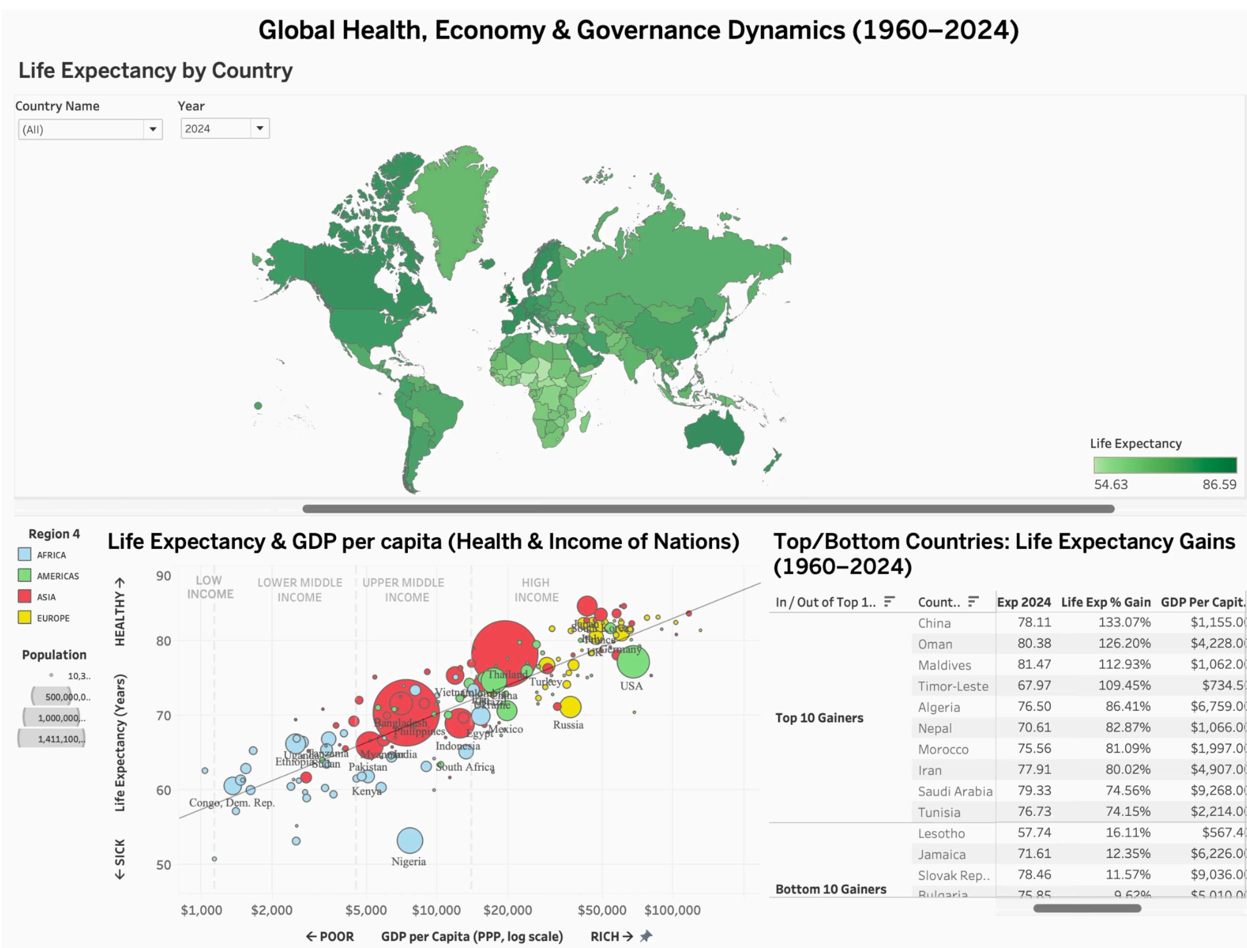
This heat map presented the top 20 countries by life expectancy alongside three governance indicators from Dataset B: Electoral Democracy (V2X Polyarchy), Liberal Democracy (V2X Libdem), and Political Corruption (V2X Corr). Color intensity on the democracy column reflected governance quality, with darker shading indicating higher democratic scores. Countries were grouped by region and sorted by life expectancy within each group. An interactive year slider enabled temporal exploration.

This chart addressed Q2 (do democratic institutions improve health outcomes?). Among the top 20 healthiest countries, most scored above 0.80 on the Electoral Democracy Index, suggesting a positive association between democratic governance and health. This finding was consistent with Montez et al., (2023), whose systematic review found that over 80% of cross-national studies documented a positive association between democracy and health outcomes even after controlling for income and education. A Stanford policy brief further confirmed that democracies consistently outperform autocracies on life expectancy and disease control, with particularly strong effects for conditions requiring sustained policy commitment (Freeman Spogli Institute, 2026).

However, notable exceptions existed: Hong Kong (0.26 polyarchy) and Singapore (0.37) achieved top-tier life expectancy despite low democracy scores. These outliers shared a common trait, they were high-income economies, suggesting that at sufficient wealth levels, effective governance can deliver health outcomes comparable to democracies regardless of electoral openness. The corruption column revealed a clearer pattern: virtually all top-performing countries had corruption scores below 0.20, regardless of their democracy scores. This suggested that low corruption might be a more consistent predictor of health outcomes than electoral democracy alone, aligning with Salazar et al., (2023) finding that reduced inequalities in premature mortality were most strongly associated with the liberal democracy index, which captures rule of law and institutional checks, rather than elections alone.

3.7 Interactive Dashboard: Unified Exploration

Overview First, Then Zoom and Filter



The interactive dashboard combined the world map, GDP scatter plot, and Top/Bottom country table into a unified view. A Year dropdown filter allowed the reader to explore any year from 1960 to 2024, with changes propagating across all three visualizations simultaneously. A Country Name filter enabled focused analysis on specific nations. The map also functioned as a cross-filter, clicking a country highlighted its position in the scatter plot and table below. This design followed the visual information-seeking mantra: the map provided overview, the filters enabled zoom, and the tooltips delivered details on demand (Shneiderman, 1996).

4. Summary and Conclusions

This project demonstrated that the relationship between wealth, governance, and health is neither simple nor linear. Economic growth translated to longer lives but with diminishing returns above approximately $15,000 GDP per capita. Democratic institutions correlated positively with health outcomes, though low corruption proved a more consistent predictor than elections alone, wealthy autocracies like Singapore achieved top-tier life expectancy while corrupt democracies lagged behind. The countries that gained the most since 1960 shared extremely low starting points rather than any particular regime type, and major crises like HIV/AIDS and COVID-19 left visible scars across all governance systems. Limitations included the ecological fallacy inherent in country-level analysis, the inability to establish causality from observational data, and data gaps for smaller territories. Nonetheless, our visualizations made these patterns accessible in ways that summary statistics alone could not, reinforcing Munzner's (2014) core argument that visualization exists to help people accomplish tasks more effectively through interactive visual representations of abstract data.

6. Contributions

Team Member	Contributions
Tambudzai Charumbira	Data collection (Gapminder, World Bank); Dataset A preparation in Tableau Prep; Tableau Desktop visualizations (Map, Bar, Line, Scatter); Dashboard design and interactivity; FIXED LOD calculated fields; report writing (Introduction, Data Story, Conclusions)
Sayan Patra	Data collection (V-Dem v16); Dataset B preparation in Tableau Prep; Tableau Desktop visualizations (Table, Heatmap, Country Rankings); Dataset relationship and extract configuration; report writing (Abstract, Datasets, References); literature review and APA citations

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